

lower.

15 A Wavelength is 8.00 cm.

The wave profile travelled by 3.20 cm in 0.20 seconds, the speed is $0.032/0.2 = 0.16 \text{ m s}^{-1}$.

Distance between A and B is 2 cm. Hence, the phase difference must be

$$\frac{\Delta x}{\lambda} = \frac{\Delta \phi}{2\pi} \rightarrow \frac{2.00 \text{ cm}}{8.00 \text{ cm}} = \frac{\Delta \phi}{2\pi} \rightarrow \Delta \phi = \pi / 2$$

16 A

$$3 \times 10^8$$